Patch-token gap-filler (arm A, io3_ptok4)

chl_cnn/ptok.py PatchTokenNet + chl_cnn/models.py DilatedCNN. train_pinn.py --model temporal3d --t3d-stem patch --ptok-patch 4 --ptok-dim 64 --ptok-heads 4 --ptok-layers 2 --ptok-arm interp --ptok-value own --context 8 --batch 8

B = batch, T = 17 days (t−8 … t+8), H × W = 149 × 181 (0.25° grid), patch grid 38 × 46 (P = 4 px), D = 64. Parameters 807,141: stem 201,762, backbone 605,379. 80 s / epoch at batch 8, 9.7 GB peak GPU memory.

patch grid 38 × 46

full resolution 149 × 181

Arrows

observed CHL pixels, full resolution

patch values, bilinear upsampling

other inputs and intermediate tensors

day t marker

1 Inputs per day

d = t−8 … t+8
9 physics fields (z)
ln CHL (z), 0 where not visible
visible mask
(B, 11, 17, H, W)
+ DOY climatology z of each day (B, 17, H, W)

context days carry the donor's clouds

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t−8 t t+8

static maps, day t (B, 7, H, W): clim z, lat, lon, log10 depth, DOY sin/cos, year

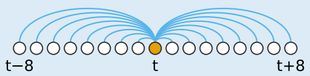
mask-aware pooling

2 Patch tokens



4 × 4 px
pad to 152 × 184, 4 × 4 px patches: 38 × 46 grid, 1,164 ocean patches per patch and day, 12 stats: CHL-anomaly mean, std (visible px), visible fraction, 9 physics means
MLP 12 → 64 → 64
+ day-offset + patch-position emb.
→ (B, 1,164, 17, 64)
fully cloudy patch: physics-only token

3 Temporal self-attention



within each patch, over its 17 day tokens; no attention across patches
2 pre-norm blocks, 4 heads, D = 64, fused SDPA, batched over patches: (B · 1,164, 17, 64)
days missing from the store masked out of the keys
→ LayerNorm → (B, 1,164, 17, 64)

4 Patch day score

$q = W_q\ \mathrm{tok(patch, t)}$
$k = W_k\ \mathrm{tok(patch, d)}$
$s = q \cdot k / \sqrt{64}$
(B, 1,164, 17) → grid
(B, 17, 38, 46)

day t's token tok(patch, t)

Fine detail (below the 4 × 4 px patch) enters only through observed full-resolution pixels: the blend values a_d (6) and day t's own channels (7).

Tokens are patch statistics; their scores (4) and FiLM (7) reach full resolution only by bilinear upsampling.

7 FiLM from day t's token

tok(patch, t) → Linear 64 → 128 (zero-initialised)
→ (γ, β) per patch
bilinear upsampling → (B, 128, H, W)
applied after the first conv: h·(1 + γ) + β

bilinear upsampling between ocean-patch centres (land excluded, normalised)
→ (B, 17, H, W)

+ per-pixel term

1×1-conv MLP 4 → 16 → 1 on visible$_d$, |d−t|/8, (d−t)/8, |SST$_d$ − SST$_t$|

per-pixel features: visible$_d$, SST$_d$, SST$_t$

each context day's own visible CHL anomaly a_d

day t's own channels (B, 11, H, W): physics, visible CHL z, visible mask

static maps (B, 7, H, W)

5 Per-pixel softmax

over context days d ≠ t visible at p + learned "no day" logit

w: (B, 17 + 1, H, W)

6 Blend

$\mathrm{blend} = \sum_d w_d(p)\, a_d(p)$

a_d: own visible CHL anomaly on day d

mass = 1 − w$_{none}$
entropy = −Σ w log w

3 × (B, 1, H, W)

7 Backbone input

concat at full res:

blend, mass, entropy (3)

day t's own channels (11)

static (7)

→ (B, 21, H, W)

8 Dilated backbone

Conv 21 → 64, 3×3
then h·(1 + γ) + β
8 residual blocks, width 64
x + [BN, GELU, Conv 3×3, dilation d] × 2
d = 1,2,4,8,16,1,2,4
BN, GELU, Conv 64 → 3, 1×1

output (B, 3, H, W): z of t−1, t, t+1
+ DOY clim z (residual)
× sd + mean → ln CHL

loss: L1 day-t hidden px
+ 0.1 × visible px
+ 100 × gradient term
+ 1 × L1 t±1 donor-hidden

For comparison: previous per-pixel attention stem (chl_cnn/tattn.py, run io2_tattn8)

per-day stack (B, 11, 17, H, W) all at full res

per-day conv embedding at full res: Conv 3×3 11 → 32, GroupNorm, GELU, Conv 3×3; + day-offset emb. (B · 17, 32, H, W)

1 self-attention block, 4 heads, dim 32, over 17 days per pixel (B · 149 · 181, 17, 32)

attention pooling per pixel: query = learned + day t's token keys / values = all 17 days, day t included → (B, 32, H, W)

+ static (7) → 39 ch → same dilated backbone

	arm A (io3_ptok4)	old stem (io2_tattn8)
stem parameters	201,762	25,984
backbone parameters	605,379	615,747
total parameters	807,141	641,731
batch	8	4
s / epoch	80	286
peak GPU memory	9.7 GB	11.3 GB